Shichao Sun

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Education

Sep. 2017 – Jan. 2020(expected)
Master of Engineering, Fudan University, China
Major in Computer Technology
GPA: 3.43/4.0

Sep. 2013 – Jun. 2017
Bachelor of Management, Fudan University, China
Major in Secrecy Policy and Management
GPA: 3.15/4.0

Research Publications

1. Xie, Z. & **Sun**, **S.** (2019, July). A goal-driven tree-structured neural model for math word problems. In *Proceedings of the twenty-eighth international joint conference on artificial intelligence, IJCAI-19* (pp. 5299–5305). International Joint Conferences on Artificial Intelligence Organization. doi:10.24963/ijcai.2019/736
2. **Sun**, **S.** & Xie, Z. (2017). Bilstm-based models for metaphor detection. In *National CCF Conference on Natural Language Processing and Chinese Computing (NLPCC 2017)*.

Awards

2019 **National Scholarship**.
2018 **Excellent Student** at Fudan University.
Scholarship for Outstanding Students at Fudan University.
2017 **Outstanding Paper** of CCF NLPCC 2017.
2014, 2015, 2016 **National Encouragement Scholarship**.

Skills

Language: Native Chinese and fluent English.
Coding: Python, LaTeX
Deep Learning Framework: PyTorch

Research Experience

2017 – 2019

Math Word Problems Solving

Input a math word problem(involving in addition, subtraction, multiplication, division, power), output the solution expression.

- Considering the fact that mathematical meaning of the number is more important than its exact value, design dynamic embedding for numbers.
- Propose a goal-driven tree-structured model to directly generate an expression tree instead of generating solution expression, motivated by the human-like method of solving math word problems.
- Source code is available at: `https://github.com/ShichaoSun/math_seq2tree`, and the paper is published in IJCAI 2019.
- Data Augmentation for solution expression by generating equivalent expressions, motivated by making the model learn math knowledge about commutative property.

2016 – 2017

Metaphor Detection

Classify whether a sentence contains the verb metaphor(like “ignite passion”).

- Model sentence by BiLSTM from word embedding to avoid using handcrafted feature.
- Design three different BiLSTM-based models motivated by three most common types of metaphor.
- Propose an ensemble model of above three models.
- Source code is available at: `https://github.com/ShichaoSun/metaphor`, and the paper is published in NLPCC 2017.

2015 – 2016

Chinese Word Embedding

Adopting word2vec algorithm to train Chinese Word Embedding and optimize it by constructing a better bag of words from dependency tree.

- Code is available at: `https://github.com/ShichaoSun/DepWord2vec`.